

ECE-UY 3613 2024 Spring: Midterm Exam

Name:

NetID:

- Write your answers **CLEARLY** and **INTELLIGIBLY** on the front pages only;
- Write your NAME and ID on the exam book;

1) (10 points) Answer the following questions regarding the Internet protocol stack:

- (4 points) How many layers does it have? What are the layers?
- (2 points) Do all the devices on the Internet have all the layers? Briefly explain why.
- (2 points) If one layer of the protocol stack has to be changed, do the layers below and above it have to be changed as well?
- (2 points) Which layer of the protocol stack has the least diversity?

Answer: a) Five Layers: Application, Transport, Network, Link, and Physical.

b) No, Devices only need layers up to their function domains. For example, routers don't need Application/Transport Layers.

c) No, Layered protocols follow modular design principle. One can change one layer without changing other layers.

d) Network layer, all devices have to use the IP protocol for inter-connectivity.

2) (10 points) Answer the following questions regarding the HTTP protocol:

- (3 points) To download a webpage with four embedded images, if HTTP 1.0 is used, how many TCP connections will be opened? if HTTP 1.1 is used, how many TCP connections will be opened?
- (3 points) HTTP itself is stateless, what mechanism does HTTP server use to keep track of client transaction history?
- (4 points) If an organization installs a HTTP proxy at the entrance of its network, what is the traffic impact on its Internet access link? what is the traffic impact on its internal network? what is the HTTP performance impact on its local users?

Answer: a) HTTP 1.0: Five connections, one for the base html file, the other four for four embedded images; HTTP 1.1: One connection to download the base html file, as well as the four images

b) HTTP uses Cookies to maintain user history.

c) Proxy can serve local users with cached web objects, so it will reduce the traffic on its Internet access link. Meanwhile, proxy will not reduce the total HTTP requests generated by the local users, so it won't decrease the volume of the internal web traffic (even though web traffic will be concentrated on the proxy server now). If web objects are served from proxy, users will experience shorter response time.

3) (10 points) Consider a home network that has only one public IP address and connects to the Internet through a wireless router. Inside the home network there is a SSH server with IP address 192.168.1.5. Consider the following protocols: HTTP, SSH, SMTP, DHCP, DNS, NAT, point out which protocol(s) are used in each of the following sequence of steps, briefly explain why.

- (2 points) Power on a laptop and connect it to the home network to get an IP address;

Answer: DHCP

- (2 points) On the laptop, type in "ssh 192.168.1.5";

Answer: SSH

- (3 points) On the laptop, type in "www.nyu.edu";

Answer: DNS, HTTP, NAT

- (3 points) On the laptop, using an email client to write an email to "someone@gmail.com".

Answer: DNS, SMTP, NAT

- 4) (12 points) Please answer the following questions regarding reliable data transfer.
- a) (2 points) Why rdt 2.0 is not sufficient to deal with channel bit corruptions?
 - b) (4 points) For rdt 2.1 to work, why does the sender have to assign a sequence number (either 0 or 1) to the packet it sends out?
 - c) (4 points) For rdt 2.1 to work, does the receiver need to assign a sequence number to ACK/NAK it sends out? Briefly explain why or why not?
 - d) (2 points) Can rdt 2.1 correctly handle packet losses?

Answer:

- a) rdt2.0 cannot handle corrupted ack/nack
- b) if ack/nack is corrupted, the sender needs to retransmit the current packet, when the receiver receives the retransmitted packet, it needs to know whether this is a new packet or a retransmission, so the sender needs to use one bit to distinguish the current packet and the next packet.
- c) Since there is no packet loss, the ACK/NAK is always for the packet the sender just sent out, there is no need to add the sequence number to it.
- d) No, if the packet or the ACK/NAK is lost, the sender and receiver will be dead-locked.

- 5) (12 points) 100 users are interested in downloading a file with 100M bytes, suppose the file server's upload bandwidth is 10Mbps (bits-per-second), each user has a download bandwidth of 10Mbps,
- (3 points) If all users download directly from the server, how long does it take for all users to finish the download of the file?
 - (5 points) If all peers have upload bandwidth of 5Mbps, and they form a P2P file sharing network to download the file in P2P fashion, how long does it take for all peers to finish the download of the file?
 - (4 points) Suppose another type of peers with 5Mbps download bandwidth and zero upload bandwidth also join the previous 100 users in the P2P file sharing network. We call this new type of peers free-riders. What is the maximum number of free-riders that the system can take before the download finish time on all peers (including the new peers) become twice as much as your answer in question b)?

Answer:

- a) the total download demand is $100 * 100M * 8$ bits, the server upload capacity is 10Mbps, so the total time for the server to complete the uploading is $100 * 100M * 8 / 10M = 8,000$ seconds; the time it takes for the slowest peer to download the whole file is $100M * 8 / 10M = 80$ seconds; So the final answer is 8,000 seconds.
- b) the time it takes for the server to send out one copy is $100M * 8 / 10M = 80$ seconds; the time it takes for the slowest peer to download the whole file is $100M * 8 / 10M = 80$ seconds; the total upload capacity in the system is $10M + 100 * 5M = 510M$, the total download time is $100 * 100M * 8 / 510M = 800 / 5.1 = 156.87$ seconds; the final completion time is $\max(80, 80, 133.3) = 156.87$ seconds.
- c) The new complete time becomes 313.74 seconds, which is longer than the time the server sends out one copy, and the time for the slowest peer to finish download. Since the total upload capacity in the system does not change after the free-riders join, that means the total download demand doubles, i.e., the number of free-riders is the same as the existing peers, that is 100.
- 6) (12 points) If host A is connected to router B, which is further connected to host C. If the link transmission rate from A to B is 10Mbps-per-second, the link transmission rate from B to C is 100Mbps-per-second, and the propagation delays on link AB and BC are both 20 milli-seconds, assuming there are no other packets in the network.
- (4 points) How long does it take to send a packet with size of 1 kilo-bytes from A to C?
 - (4 points) If A uses stop-and-wait transport protocol to send data to B reliably, what are the link utilizations on AB and BC, respectively? You can ignore the transmission delay of ACK/NAK packets, and the propagation delays on each link is symmetric on both directions.
 - (4 points) If A uses a sliding-window based transport protocol to send data to B reliably, what is the highest throughput A can achieve? what is the minimum window size to achieve the highest throughput? what are the link utilizations on AB and BC, respectively?

Answer: a) packet transmission time on AB = $1,000 * 8 / 10M = 0.8$ ms, on BC = $1,000 * 8 / 100M = 0.08$ ms; the total time = $0.8 + 0.08 + 20 * 2 = 40.88$ ms;

- b) the round-trip time is $40.88 + 40 = 80.88$ ms, with stop-and-wait, one packet is transmitted on each link every one round trip time, the throughput is $8,000 / 80.88ms \sim 0.1Mbps$, the utilization on link AB is $0.1Mbps / 10Mbps = 0.01$; on link BC is $0.1Mbps / 100Mbps = 0.001$;
- c) the end-to-end bottleneck link is AB with rate of 10Mbps, the highest end-to-end throughput is 10Mbps; the minimum window size is $10M / 0.1M = 100$, the utilization on AB will be 100%, on BC will be 10%.
- 7) (10 points) Given an IP address 128.119.174.79, please find
- (3 points) the class of the network that this address belongs to (Class A, B, C, D or E), and how many hosts can be placed into this network?
 - (4 points) if we want to evenly divide this network into 16 subnets, what would be the subnet mask for the subnets? how many hosts can be placed into each subnet?
 - (3 points) what is the subnet ID of the subnet that IP address 128.119.174.79 belongs to? what is the host part of this address in its subnet?

Answer: Binary IP address is: 1000 0000. 0111 0111. 1010 1110. 0100 1111

- a) Class B (first two bits of IP address is 10), the last 16 bits are used for host IDs, we can have $2^{16} - 2 = 65,534$ hosts.
- b) To create 16 subnets, we add the first 4 bits in the host part to subnet mask, the subnet mask is 1111 1111. 1111 1111. 1111 0000. 0000 0000, the number of hosts in each subnet is $2^{12} - 2 = 4,094$
- c) The subnet ID is the first 20 bits of the IP address, it is 10000000.01110111.10100000.00000000; The host part is the last 12 bits: 1110 0100 1111, or (3663 decimal)
- 8) (12 points) A TCP connection will experience a time-out packet loss whenever the sender's congestion window reaches 128 segments, and suppose there is no other packet loss in the network, if the round-trip-time of this connection is 200 millisecond, and the packet size is 1,500 bytes
- a) (6 points) how often does this TCP connection experience time-out packet losses? In other words, what is the time interval between two consecutive packet losses?
- b) (6 points) what is the average sending rate (measured in bits-per-second) of the TCP sender in long run?

Answer:

- a) after each time-out loss, TCP will do slow-start, set the ssthresh to 64 segments, reduce cwnd=1 segment, double it every round-trip time, until it reaches 64 segments in the 7-th round, then increase it by 1 segment every RTT, until it reaches 128 segments, the total time is $7+64=71$ RTTs= $71*200\ ms=14.2$ seconds.
- b) average window size is $(1 + 2 + 4 + 8 + 16 + 32 + 64 + \sum_{i=65}^{128} i)/71 = (63 + 65 * 96)/71 = 6303/71$, the average sending rate:

$$\frac{6303 * 1500 * 8}{71 * 0.2} = 5.326 Mbps$$

- 9) (12 points) At time zero, a TCP sender begins to send out a total of five segments to a receiver, one segment every 5 ms, and each segment contains 200 bytes of data, and the initial sequence number on the sender is 150. Suppose the receiver sends back one acknowledgement for each received segment. Assume the second segment is lost, and there is no other segment loss, the one-way delay between the sender and receiver is 50 ms, and the time-out value is 200ms, you can ignore the processing delay on the receiver side. Plot the packet exchanges between the sender and receiver over time till all the five segments are received. (assuming TCP uses go-back-N).

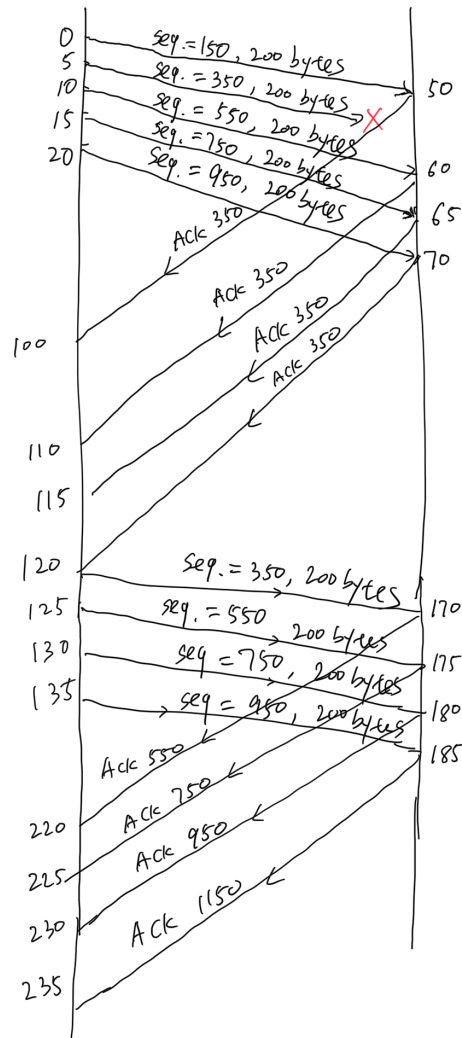


Fig. 1. TCP Packet Exchange

If you use up the space under any particular problem, you can write your answer here. On the page of the problem, tell us part of your work is here and write down the page number of this page and. Don't tear off any pages.

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